

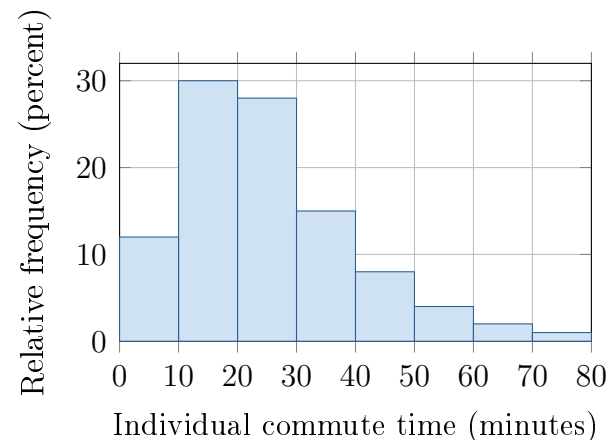
When Is the Mean Nearly Normal?

Unit 2 · Topic 2.12 · THE PROBLEM

Individual commute times in a very large city are strongly right-skewed, with population mean $\mu = 25$ minutes and standard deviation $\sigma = 15$ minutes. Consider independent random samples.

Rae: “Any average is approximately normal, even for $n = 4$.”

Sol: “With this long right tail, $n = 4$ may be too small; $n = 40$ is more defensible.”



FIRST TAKE — one sentence before you work the parts

Which would you expect to smooth out the long commute times more: averaging 4 people or 40? Give one reason.

- DOK 1** (a) Identify the population shape and state the population mean and standard deviation with units.
- DOK 2** (b) For $n = 4$ and $n = 40$, calculate the mean and standard deviation of the sampling distribution of $\bar{x}$.
- DOK 3** (c) In 3–4 sentences, decide whose claim is better supported. Use the population shape, independence, and the two sample sizes. Explain which distribution the CLT describes and why using a normal curve for averages of 4 could be unreliable.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.